# Daniel Peelen

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## Education

**University of Pittsburgh** September 2023 - May 2027

*B.S. Computer Science, Hungarian Minor — 3.9 GPA, Dean's List*

- Coursework: Deep Learning, Data-Driven Systems, Data Structures & Algorithms, Systems Software

## Experience

**Software Engineering Intern** May 2026 – August 2026

*Microsoft* *Redmond, WA*

- Analyzed 1,000,000+ Word Copilot crashes with Kusto and OpenTelemetry by building an LLM reliability agent.
- Shipped an **org-wide plug-in for Microsoft's internal Copilot**, giving engineers on-demand crash analyses.
- Fixed Word bugs surfaced by agent's output, **eliminated 50,000+ enterprise-facing crashes over 2 releases**.

**Software Engineer Intern** January 2026 – April 2026

*Synopsys* *Pittsburgh, PA*

- Designed and implemented Ansys Mechanical's RAG coding agent memory solution with semantic clustering.
- Built chatbot's full test suite and a multi-turn eval harness with bucket visualization on Ansys test infrastructure.
- **Raised multi-turn accuracy from 40% to 82%** by shipping Go code to production behind feature flags.

**Software Engineer Intern** May 2025 – August 2025

*PNC Financial Services* *Pittsburgh, PA*

- Built a full-stack **housing analytics app** that turned **1M+ housing records** into homeowner-facing insights.
- Authored Angular unit tests, raising coverage from 45% to 85% and cutting production bugs by 35%

**Teaching Assistant** August 2024 – April 2025

*University of Pittsburgh* *Pittsburgh, PA*

- Led recitations and practices for 100+ students in Discrete Math, **raised average final scores from 78 to 84.**
- Held 5+ hours of weekly office hours, coaching students on **proof techniques and algorithmic thinking**.

## Projects

**DannyTorch** | *Python, NumPy*

- Designed deep learning framework in pure NumPy **without AI**, building an autograd engine over a dynamic graph
- Implemented full transformer stack (MHA with causal masking, RoPE, RMSNorm, SwiGLU) behind a module API
- Verified all autograd ops against gradients, then trained a small LM on one A100, **cutting loss from 4.9 to 1.2**

**GhostPIN** | *Node.js, Redis, PostgreSQL, WebAuthn* **2nd Place, HackHarvard 2025**

- Designed an anti-phishing SDK with RSA-signed protocol and origin-bound nonces that block fake checkout pages.
- Integrated biometric MFA into checkout flow via WebAuthn, with approvals verified end-to-end in under 200ms.

**Mantis** | *Python, OpenCV, YOLOv8, GCP, Twilio* **2nd place, CalHacks 2025**

- Built a full-stack surveillance platform detecting theft and weapons in live feeds with real-time owner alerts.
- Architected a Python detection core streaming over WebSockets into a reactive Next.js/Convex dashboard on GCP.

## Technical Skills

**Languages**: Python, C/C++, Java, Go, TypeScript, JavaScript, SQL, OCaml
**Machine Learning & AI**: PyTorch, NumPy, HuggingFace, transformers, model evals, benchmarking
**Cloud & DevOps**: AWS, GCP, Azure DevOps, Docker, Linux, Git
**Backend & Frontend**: Spring Boot, Kafka, Redis, React, Angular

## Leadership

**Director** January 2025 – September 2025

*SteelHacks* *Pittsburgh, PA*

- Grew the event **21%** year over year to 600+ attendees and raised $50K+ from Google, Amazon, and PNC.

**Technical Team Member** September 2026 – Present

*Google Developer Group, Budapest University of Technology and Economics* *Budapest, Hungary*

- Led team of engineers to build myBME for students and co-organize guest tech lectures **while studying abroad.**